

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018



Mini Project Report on

“Smart Fire and Smoke Detection and Targeted Suppression System”

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards the completion of the
Mini Project under the **Inter Disciplinary Project Based Learning**,
Department of Basic Sciences.

by

SN	Name	USN/Roll number
1	Varun R K	1CR25CS205
2	Vedhamrutha Sathi	1CR25CS206
3	Samiksh M	1CR25EC185
4	Sandeep S	1CR25EC186

Under the Guidance of
Dr. Ramdas Balan

Asst. Professor, Head, Basic Science Research Centre Dept. of Physics



CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI,
BANGALORE-560037

CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI,
BANGALORE-560037

DEPARTMENT OF BASIC SCIENCES



CERTIFICATE

This is to certify that the File Structures mini project entitled “Smart Fire and Smoke Detection and Targeted Suppression System” has been successfully carried out by Varun R K (1CR25CS205), Vedhamrutha Sathi (1CR25CS206), Samiksh M (1CR25EC185), and Sandeep S (1CR25EC186), Bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Inter Disciplinary Project Based Learning, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

Signature of Guide

Dr. Ramdas Balan
Asst. Professor
Dept. of Physics, CMRIT

Signature of HOD

Dr. Raveesha K H
HoD,
Dept. of Physics, CMRIT

External Viva

Name of the examiners

1.

2.

Signature with date

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Varun R K (1CR25CS205),

Vedhamrutha Sathi (1CR25CS206),

Samiksh M (1CR25EC185), and

Sandeep S (1CR25EC186)

Abstract

The Smart Fire Detection and Targeted Suppression System is designed to improve fire safety by providing accurate fire detection and efficient emergency response. Traditional fire detection systems often depend on a single sensor, which can result in false alarms and unnecessary water wastage. The proposed system uses an Arduino-based multi-sensor approach that combines flame, smoke, and temperature sensors to detect fire incidents more reliably. The system is also capable of differentiating between fake flames and real fire conditions, thereby reducing false detections. Once a real fire is identified, a relay-controlled water pump activates a sprinkler mechanism only in the affected area, ensuring targeted suppression and minimizing water consumption. Additional alert mechanisms such as buzzers are included to provide immediate warning during emergencies. The prototype was tested under different conditions and demonstrated quick response time, accurate fire identification, and effective localized suppression. The developed system offers a cost-effective, reliable, and intelligent solution suitable for homes, offices, laboratories, and industrial environments to enhance overall fire safety and prevention.

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Chapter 1

INTRODUCTION

Fire accidents in homes, industries, and public places can cause serious damage to life and property. Traditional fire detection systems mainly rely on a single sensor, which often leads to false alarms and delayed response. To overcome these limitations, this project proposes a Smart Fire Detection and Targeted Suppression System using Arduino. The system uses multiple sensors such as flame, smoke, and temperature sensors to accurately detect real fire incidents. It also differentiates between fake flames and real flames to reduce false alarms. Once fire is confirmed, the system activates water sprinkling only in the affected area, preventing unnecessary water wastage. This project improves safety, response time, and efficiency using automation and intelligent sensing techniques.

1.1 Brief history of Fire and Smoke system

Early fire detection systems mainly used smoke alarms and manual monitoring methods. Later, automatic systems using heat and flame sensors were introduced. However, many traditional systems still face issues like false alarms and delayed response.

1.2 Modern Fire safety system

Modern fire safety systems use IoT, sensors, and automation for faster and more accurate detection. Smart systems can send alerts, monitor conditions in real time, and activate suppression mechanisms automatically.

Chapter 2

Problem Statement

2.1 Description

Existing fire detection systems often produce false alarms because they mainly rely on a single sensor, such as a smoke or flame sensor, which may react to harmless conditions like cigarette smoke, cooking fumes, heat, or artificial light sources. These systems are unable to accurately distinguish between real fire incidents and fake flame conditions, leading to unnecessary panic and interruption. In many cases, traditional systems also activate sprinklers throughout the entire building whenever fire is detected, regardless of the exact location of the fire. This results in excessive water wastage and can cause additional damage to electrical equipment, furniture, documents, and other valuable materials. Furthermore, delayed identification of the actual fire source reduces the efficiency of emergency response systems. Therefore, there is a need for a smart and reliable fire detection system that can accurately identify real fire conditions and activate targeted suppression only in the affected area.

2.2 Challenge Statement

To design and implement a smart fire detection system that:

- Detects real fire accurately
- Differentiates fake flame from real flame
- Activates water sprinkling only in the affected area
- Reduces false alarms and water wastage
- Provides faster emergency response

2.3 Objective of the Project

- To design a smart fire detection system using Arduino.
- To detect real fire using multiple sensors.
- To reduce false alarms caused by fake flames.
- To activate water sprinkling only in the affected area.
- To improve fire safety and reduce water wastage.

2.4 Proposed Solution

The proposed system is a Smart Fire Detection and Targeted Suppression System using Arduino. It uses flame, smoke, and temperature sensors to detect fire accurately and reduce false alarms. The system can differentiate between real and fake flames and activates water sprinkling only in the affected area. A buzzer alert is also provided for emergency warning. This system improves fire safety while reducing unnecessary water wastage.

2.5 Advantages of Proposed solution

- Detects fire more accurately using multiple sensors
- Reduces false alarms by identifying real and fake flames
- Activates sprinklers only in the affected area
- Minimizes water wastage and property damage
- Provides quick automatic response during emergencies
- Low-cost and easy to implement using Arduino
- Improves overall fire safety and reliability

Chapter 3

3.1 Design Thinking Process

a) Empathize

Discussions with students and faculty identified the need for an affordable and accurate fire safety system with reduced false alarms.

b) Define

Main issues identified:

- False fire detection
- Delayed emergency response
- Excessive water wastage
- Inaccurate fire localization

c) Ideate

Different ideas were discussed including smoke-only detection, temperature-based systems, and multi-sensor systems. The final idea selected was a smart multi-sensor fire detection and targeted suppression system.

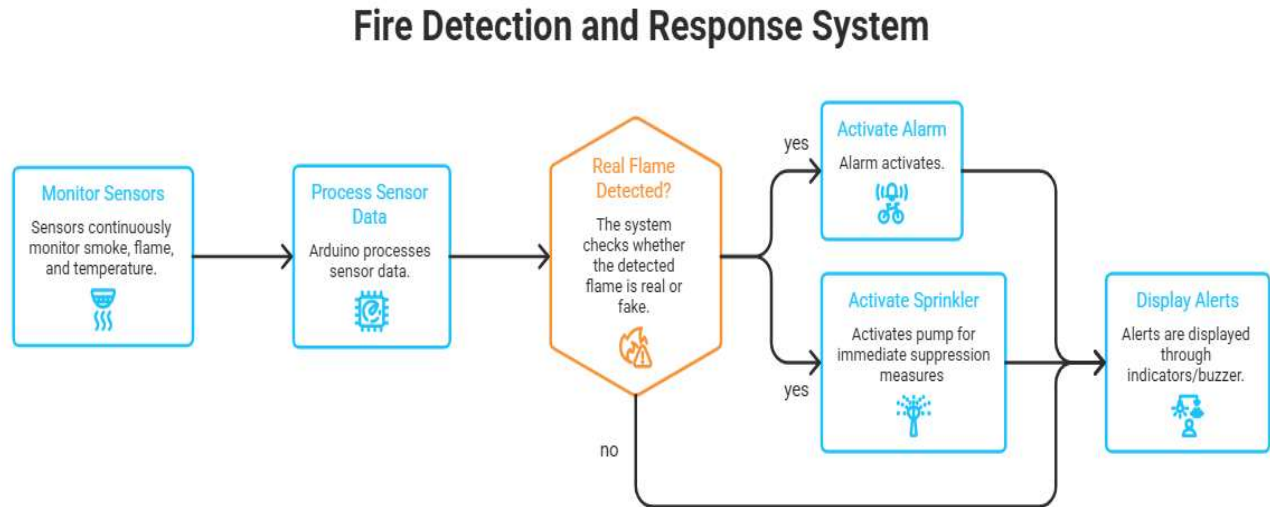
d) Prototype

A prototype was developed using Arduino, flame sensor, smoke sensor, temperature sensor, water pump, and buzzer.

e) Test

The system was tested under different conditions and successfully detected real fire while reducing false alarms.

3.2 Methodology



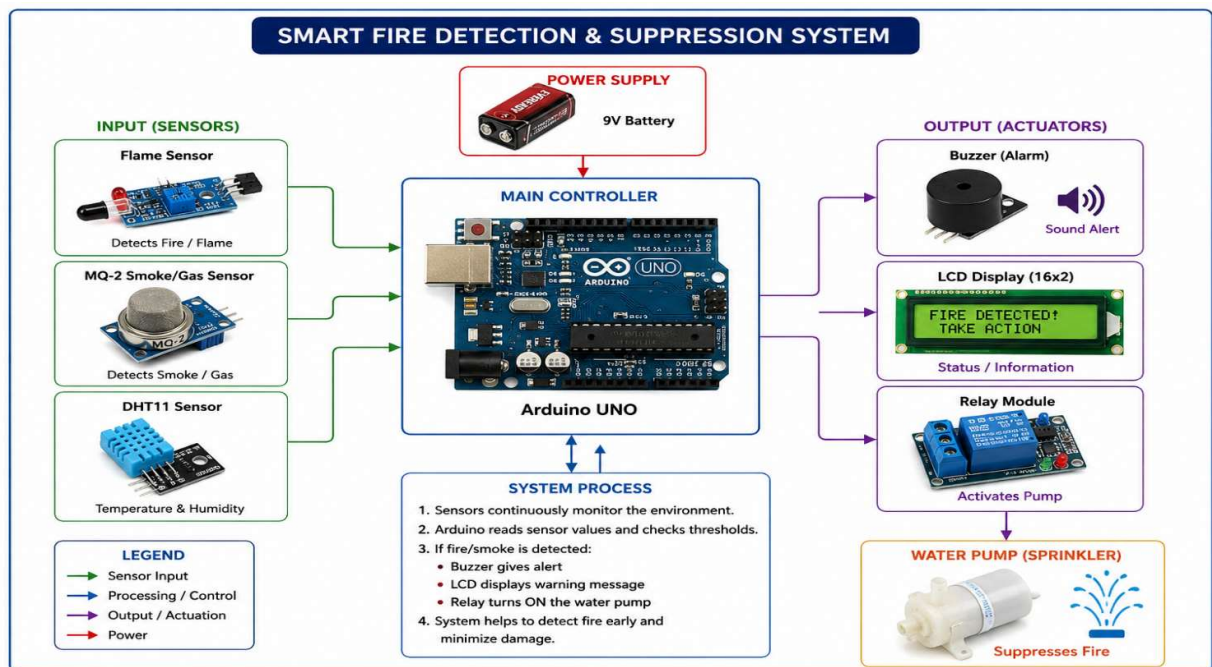
The proposed system works by continuously monitoring the surroundings using flame, smoke, and temperature sensors connected to an Arduino microcontroller. The sensor data is analysed to determine whether the detected condition represents a real fire or a false flame source. When smoke concentration, temperature, and flame intensity exceed the predefined threshold values simultaneously, the system confirms the presence of fire. Once a real fire is detected, the Arduino activates a relay-controlled water pump that sprinkles water only in the affected area, reducing unnecessary water wastage. A buzzer is also triggered to provide an immediate alert during emergencies. The entire process is automatic, enabling quick response and improving the overall efficiency and reliability of fire safety management.

3.3 Prototype Description

3.3.1 Materials Used

- Arduino Uno – used as the main controller for the system
- Flame Sensor – detects using IR to detect fire
- Smoke Sensor (MQ2) – used to detect smoke and harmful gases
- Temperature Sensor – used to detect change in temperature
- Water Pump – used for sprinkler system
- Relay Module – used for the DC motor pump
- Buzzer – used to provide an audible alert
- Wi-Fi Module (ESP32) – used for alert
- Jumper Wires
- Breadboard
- Battery/Power Supply

3.3.2 System Diagram



Chapter 4

Implementation

Arduino Code

```
1  #include <LiquidCrystal_I2C.h>
2  #include <DHT.h>
3  #include <SoftwareSerial.h>
4
5  #define GAS_SENSOR_APIN  A0
6  #define FLAME_SENSOR_DPIN  6
7  #define BUZZER_PIN      8
8  #define DHT_PIN         2
9  #define DHT_TYPE         DHT11
10 #define MOTOR_PIN       9
11
12 SoftwareSerial espSerial(10, 11); // D10=RX, D11=TX → ESP32
13
14 LiquidCrystal_I2C lcd(0x27, 16, 2);
15 DHT dht(DHT_PIN, DHT_TYPE);
16
17 void setup() {
18     Serial.begin(9600);
19     espSerial.begin(9600);
20     pinMode(FLAME_SENSOR_DPIN, INPUT);
21     pinMode(BUZZER_PIN, OUTPUT);
22     pinMode(MOTOR_PIN, OUTPUT);
23     digitalWrite(MOTOR_PIN, LOW);
24     dht.begin();
25
26     lcd.init();
27     lcd.backlight();
28     lcd.setCursor(0, 0);
29     lcd.print(" Safety System ");
30     lcd.setCursor(0, 1);
31     lcd.print(" Initializing..");
32     delay(2000);
33     lcd.clear();
34
35     tone(BUZZER_PIN, 1000, 300);
36     delay(400);
37     tone(BUZZER_PIN, 1500, 300);
38     delay(400);
39     noTone(BUZZER_PIN);
40
41     Serial.println("=====");
42     Serial.println("      Safety System Ready      ");
43     Serial.println("=====");
44     Serial.println("  GAS AO      → A0");
45     Serial.println("  FLAME DO     → D6");
46     Serial.println("  DHT DATA    → D2");
47     Serial.println("  BUZZER +     → D8");
48     Serial.println("  MOTOR/RELAY  → D9");
49     Serial.println("  LCD SDA      → A4");
50     Serial.println("  LCD SCL      → A5");
51     Serial.println("  ESP32 TX     → D10");
52     Serial.println("  ESP32 RX     → D11");
53     Serial.println("=====");
54     delay(1000);
55 }
56
57 void sendToESP(String msg) {
58     espSerial.println(msg);
59     Serial.print("Sent to ESP32: ");
60     Serial.println(msg);
61 }
62
```

ESP32

```
1 #define BLYNK_TEMPLATE_ID "TMPL31Pe2sjtP"
2 #define BLYNK_TEMPLATE_NAME "fire safety system"
3 #define BLYNK_AUTH_TOKEN "2nP1gj5PPPxbAKlLOLxrYHYd2vHBtW54" // ← Replace later
4
5 #include <WiFi.h>
6 #include <BlynkSimpleEsp32.h>
7
8 char ssid[] = "Ashborn"; // ← Replace later
9 char pass[] = "12345678"; // ← Replace later
10
11 // ESP32 Serial2 → reads from Arduino
12 // Arduino D11 (TX) → ESP32 GPIO16 (RX2)
13 // Arduino D10 (RX) → ESP32 GPIO17 (TX2)
14 // Arduino GND → ESP32 GND
15
16 void setup() {
17   Serial.begin(115200);
18   Serial2.begin(9600, SERIAL_8N1, 16, 17); // RX=GPIO16, TX=GPIO17
19
20   Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
21   Serial.println("ESP32 Ready. Listening for Arduino...");
22 }
23
24 void loop() {
25   Blynk.run();
26
27   if (Serial2.available()) {
28     String message = Serial2.readStringUntil('\n');
29     message.trim();
30
31     Serial.print("Received from Arduino: ");
32     Serial.println(message);
```

Chapter 5

Results and Analysis

User Testing & Feedback

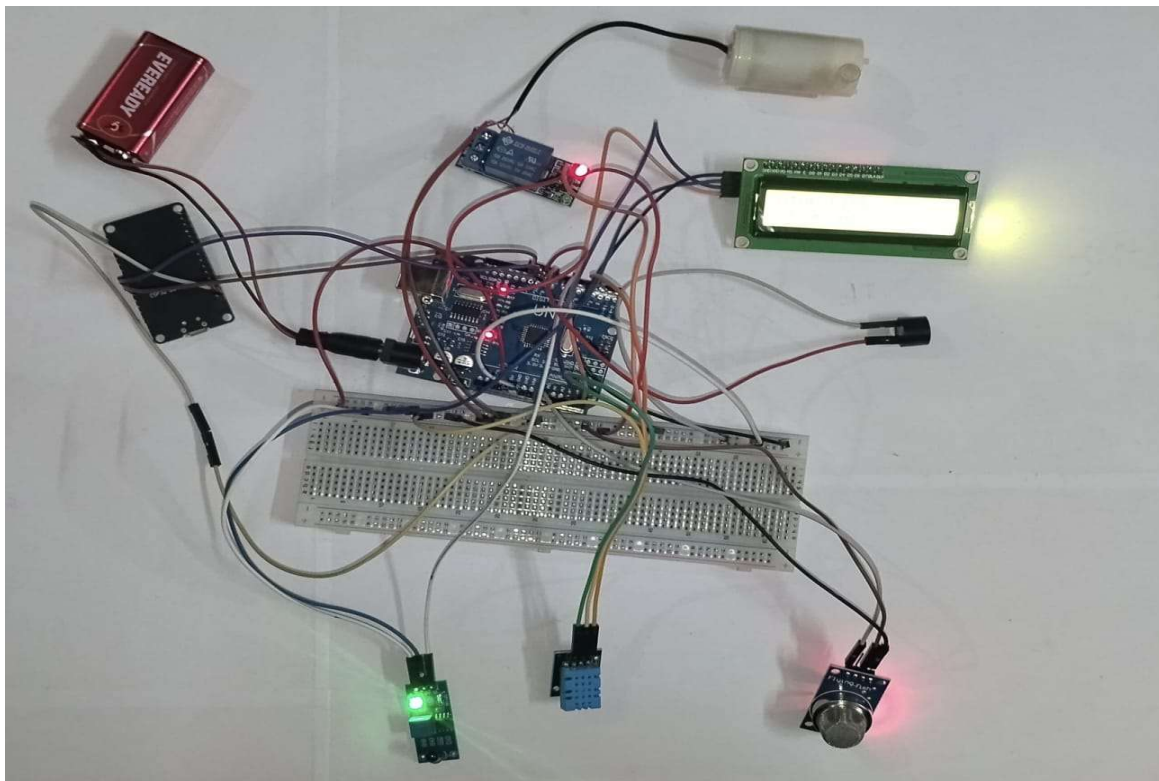
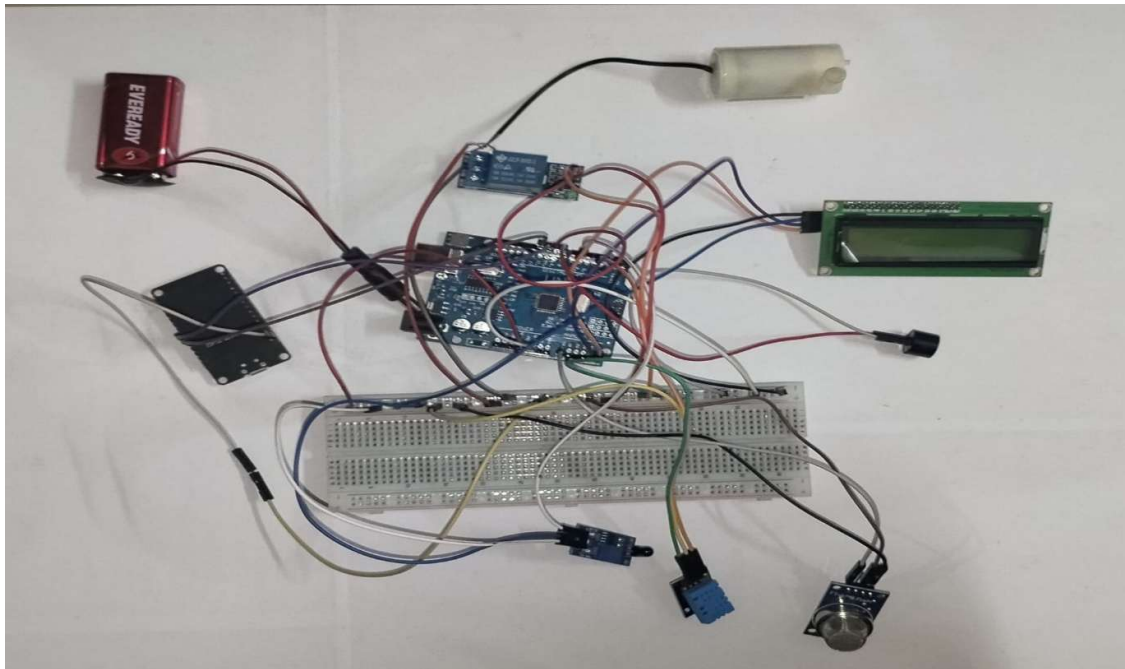
Quantitative Results

- Fire detected within a few seconds
- Reduced false alarms compared to normal flame sensors
- Water sprayed only in affected region
- Successful real-time monitoring

Qualitative Feedback

- “System responds quickly to fire.”
- “Targeted sprinkling reduces water wastage.”
- “Fake flame detection improves accuracy.”

Project Pictures



Chapter 6

Conclusion & Future Work

The Smart Fire Detection and Targeted Suppression System provide an effective and intelligent solution for fire safety. By combining multiple sensors and targeted suppression techniques, the system improves detection accuracy and minimizes false alarms and water wastage.

Future Work

- Mobile app notifications
- IoT cloud monitoring
- AI-based fire prediction
- Automatic emergency service alerts
- Camera-based fire verification

References

Research Papers

- Arduino Official Documentation
- Flame Sensor Datasheets
- MQ2 Smoke Sensor Documentation
- Research papers on Smart Fire Detection Systems
- IoT and Embedded Systems textbooks

Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles